

## Import Necessary Libraries

```
In [4]: import pandas as pd
from sklearn.cluster import KMeans
from sklearn.cluster import DBSCAN
from sklearn.metrics import silhouette_score
from sklearn.preprocessing import scale
```

## Parse Data

```
In [ ]: # parse data
df = pd.read_csv("retail_transactions.csv", encoding='iso-8859-1')
df['item_total'] = df['Quantity'] * df['UnitPrice']
dfnew = df.groupby('CustomerID').agg({'InvoiceNo': 'nunique', 'item_total': 'sum'})
dfnew.rename(columns={'InvoiceNo': 'invoice_count', 'item_total': 'amt_total'}, inplace=True)
X = scale(dfnew[['invoice_count', 'amt_total']])
```

## Create Functions for Kmeans and DbScan

```
In [8]: def cluster_kMeans(X,k):
    mdlKmcRtc = KMeans(n_clusters=k, random_state=1)
    mdlKmcRtc.fit(X)
    kmeans_labels = mdlKmcRtc.labels_
    kmeans_sil_score = silhouette_score(X, kmeans_labels)
    return kmeans_labels, kmeans_sil_score
```

```
In [ ]: def cluster_DBScan(X,e):
    mdlDbsRtc = DBSCAN(eps=e)
    mdlDbsRtc.fit(X)
    dbscan_labels_e = mdlDbsRtc.labels_
    dbscan_sil_score_e = silhouette_score(X, dbscan_labels_e)
    dbscan_clusters_e = len(set(dbscan_labels_e))-(1 if -1 in dbscan_labels_e else 0)
    dbscan_anomalies_e = len(dbscan_labels_e[dbscan_labels_e==-1])
    return dbscan_labels_e, dbscan_sil_score_e, dbscan_clusters_e, dbscan_anomalies_e
```

## Try different K and e Values

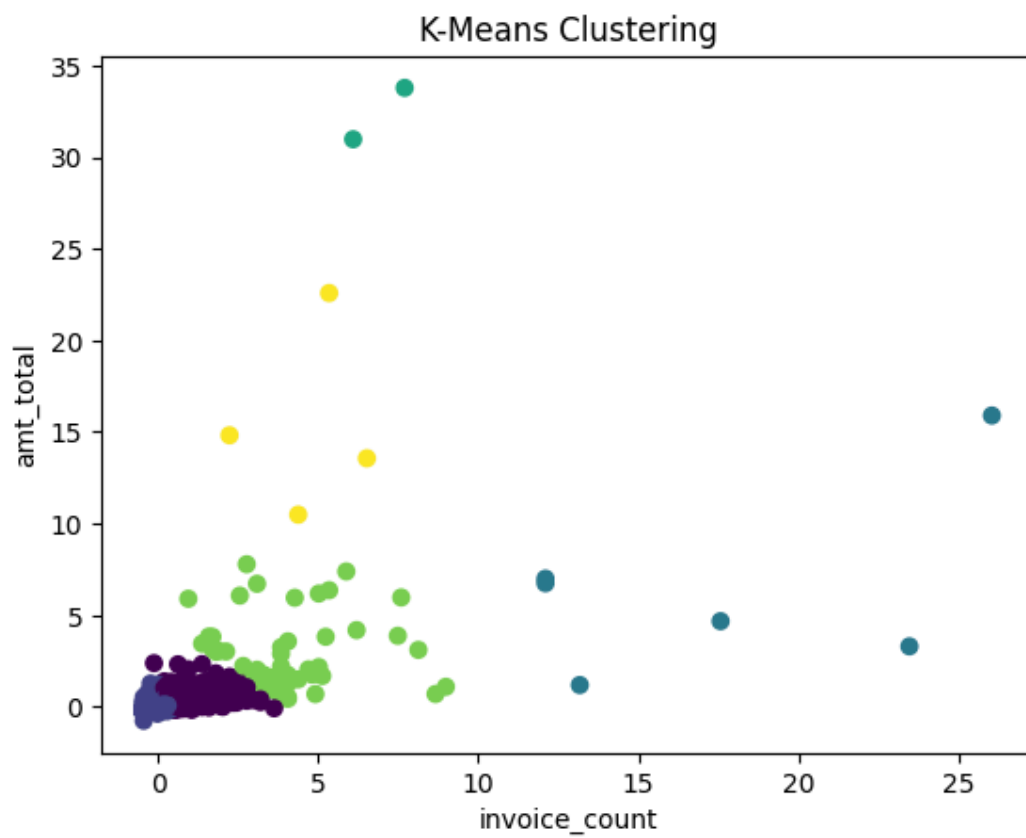
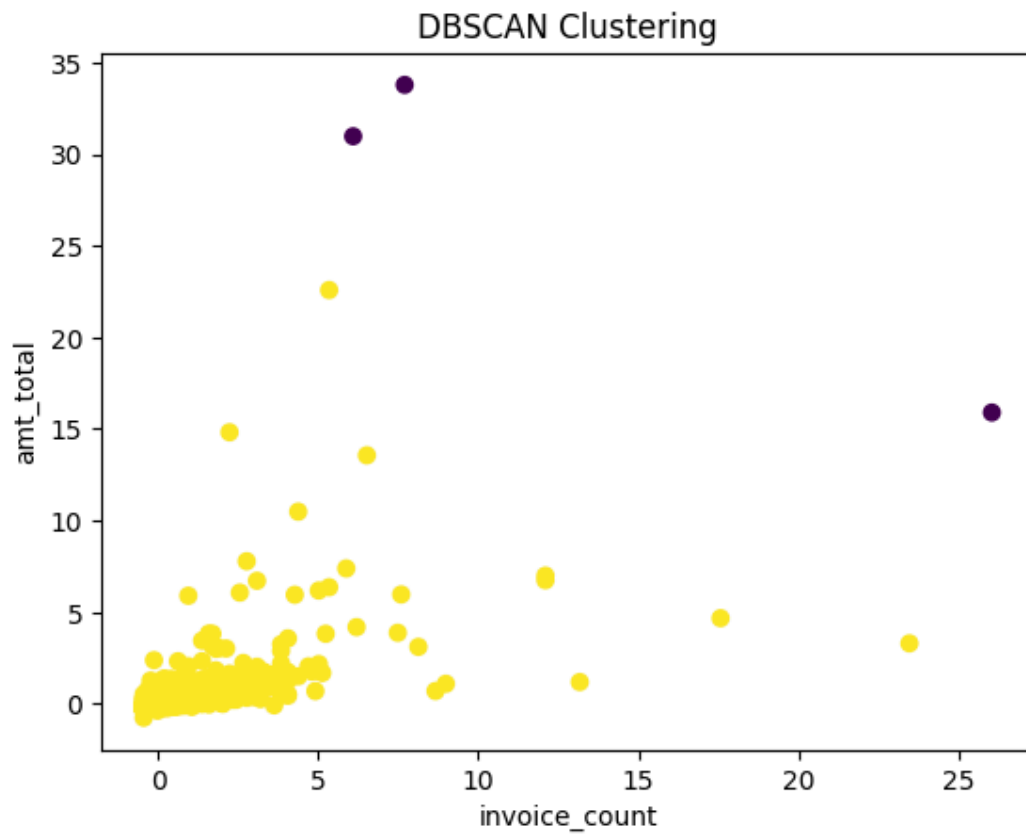
```
In [11]: for k in [2,3,4,5,6]:
    kmeans_sil_score_best = 0
    kmeans_labels_k, kmeans_sil_score_k = cluster_kMeans(X,k)
    if kmeans_sil_score_k > kmeans_sil_score_best:
        kmeans_labels = kmeans_labels_k
        kmeans_sil_score_best = kmeans_sil_score_k
    print('k:',k, '\t\t','sil_score:',kmeans_sil_score_k)

for e in [0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,1.5,2.0,2.5,3.0,4.0,5.0,6.0,7.0,8.0,9.0]:
    dbscan_sil_score_best = 0
    dbscan_labels_e, dbscan_sil_score_e, dbscan_clusters_e, dbscan_anomalies_e = cluster_DBScan(X,e)
    if dbscan_sil_score_e > dbscan_sil_score_best:
        dbscan_labels_e = dbscan_labels_e
        dbscan_sil_score_best = dbscan_sil_score_e
    print('eps:',e, '\t\t','sil_score:',dbscan_sil_score_e, '\t\t','clusters:',dbscan_clusters_e, '\t\t','anomalies:',dbscan_anomalies_e)
```

|                |                                |              |   |
|----------------|--------------------------------|--------------|---|
| k: 2           | sil_score: 0.9503432954038323  |              |   |
| k: 3           | sil_score: 0.8229586806895549  |              |   |
| k: 4           | sil_score: 0.7675493870106322  |              |   |
| k: 5           | sil_score: 0.7514378910638715  |              |   |
| k: 6           | sil_score: 0.7299224375528661  |              |   |
| eps: 0.1       | sil_score: 0.44303062146907807 | clusters: 24 |   |
| anomalies: 235 |                                |              |   |
| eps: 0.2       | sil_score: 0.8835745228080347  | clusters: 1  | a |
| nomalies: 110  |                                |              |   |
| eps: 0.3       | sil_score: 0.9082453438451032  | clusters: 1  | a |
| nomalies: 75   |                                |              |   |
| eps: 0.4       | sil_score: 0.8723010302239043  | clusters: 2  | a |
| nomalies: 60   |                                |              |   |
| eps: 0.5       | sil_score: 0.8277387470113158  | clusters: 3  | a |
| nomalies: 43   |                                |              |   |
| eps: 0.6       | sil_score: 0.8260463106415202  | clusters: 3  | a |
| nomalies: 34   |                                |              |   |
| eps: 0.7       | sil_score: 0.8336443702741331  | clusters: 2  | a |
| nomalies: 32   |                                |              |   |
| eps: 0.8       | sil_score: 0.9460932490176458  | clusters: 1  | a |
| nomalies: 30   |                                |              |   |
| eps: 0.9       | sil_score: 0.9472191707848616  | clusters: 1  | a |
| nomalies: 29   |                                |              |   |
| eps: 1.0       | sil_score: 0.9472191707848616  | clusters: 1  | a |
| nomalies: 29   |                                |              |   |
| eps: 1.5       | sil_score: 0.9502179515979182  | clusters: 1  | a |
| nomalies: 26   |                                |              |   |
| eps: 2.0       | sil_score: 0.9129211142544389  | clusters: 2  | a |
| nomalies: 17   |                                |              |   |
| eps: 2.5       | sil_score: 0.9651633796425144  | clusters: 1  | a |
| nomalies: 12   |                                |              |   |
| eps: 3.0       | sil_score: 0.9651633796425144  | clusters: 1  | a |
| nomalies: 12   |                                |              |   |
| eps: 4.0       | sil_score: 0.9670456336391249  | clusters: 1  | a |
| nomalies: 10   |                                |              |   |
| eps: 5.0       | sil_score: 0.972935604821862   | clusters: 1  | a |
| nomalies: 6    |                                |              |   |
| eps: 6.0       | sil_score: 0.9744991911017402  | clusters: 1  | a |
| nomalies: 5    |                                |              |   |
| eps: 7.0       | sil_score: 0.9755361328500473  | clusters: 1  | a |
| nomalies: 4    |                                |              |   |
| eps: 8.0       | sil_score: 0.9755361328500473  | clusters: 1  | a |
| nomalies: 4    |                                |              |   |
| eps: 9.0       | sil_score: 0.9768641056046513  | clusters: 1  | a |
| nomalies: 3    |                                |              |   |

```
In [12]: import matplotlib.pyplot as plt
plt.scatter(X[:,0], X[:,1],c=dbscan_labels)
plt.xlabel('invoice_count')
plt.ylabel('amt_total')
plt.title('DBSCAN Clustering')
plt.show()

plt.scatter(X[:,0], X[:,1],c=kmeans_labels)
plt.xlabel('invoice_count')
plt.ylabel('amt_total')
plt.title('K-Means Clustering')
plt.show()
```



In [ ]: